

Transmission Spectra in EXHALE_transit.py: H α /H β from the Non-LTE $n = 2$ Population, and the Metal Resonance Doublets

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1. Overview

EXHALE_transit.py (the Transmission Probability Model shipped with EXHALE) computes transit transmission spectra by integrating the line-of-sight (LOS) optical depth on an impact-parameter grid, using Voigt profiles evaluated through the Faddeeva function, and then averaging over the projected stellar disk with optional instrument and planet-rotation convolutions. The original code handles the He I 2³S triplet (10830 Å) and the H I / D Ly α doublet (1215.67 Å).

This memo documents the addition of the H α (6562.8 Å, $n = 2 \rightarrow n = 3$) transmission spectrum. Unlike Ly α , which absorbs out of the ground state ($n = 1$) that traces the bulk of the atomic hydrogen, H α absorbs out of the $n = 2$ level,

whose population must be computed from a non-LTE balance. We follow **Christie, Arras & Li (2013, ApJ 772, 144)**, including collisional excitation/de-excitation, recombination cascade, two-photon decay, ℓ -mixing, and **Ly α radiative pumping** ($1s \leftrightarrow 2p$), with the rate coefficients of EXHALE's own $n = 2$ model. The Ly α mean intensity $J_{\text{Ly}\alpha}(r)$ is not in the profiles the tool reads (a run with the excited-hydrogen model writes its own field to output/Excited_H.txt, which is not read); it is either read from an external text file or, by default, estimated with the simple prescription of Huang et al. (2017), see §2.2.

The H α branch always runs; only the source of $J_{\text{Ly}\alpha}$ depends on whether Jlya_file is set. The existing He/Ly α behavior is unchanged.

This memo also documents the **metal resonance doublets** (Mg II h&k, Ca II H&K, Na I D; §7.), which are computed by the same line-of-sight integration and produce the same three-curve spectrum figures as the H/He lines.

2. The $n = 2$ hydrogen population (Christie et al. 2013)

The $2s$ and $2p$ states are treated separately because they have very different lifetimes (the $2p \rightarrow 1s$ Ly α decay is fast, $A_{2p \rightarrow 1s} = 6.2649 \times 10^8 \text{ s}^{-1}$, whereas the $2s \rightarrow 1s$ two-photon decay is slow, $A_{2s \rightarrow 1s} = 8.22461 \text{ s}^{-1}$, Drake 1986, eq. 27). With $n_{1s} = n_{\text{HI}}$, n_{HII} , n_{HeII} , n_{HeIII} , n_e and T taken from the EXHALE profiles, the rate-equilibrium equations are those of Christie et al. (Eqs. 12–13) as the solver writes them (excited_hydrogen.f90, n2_rate_matrix):

$$[A_{2p \rightarrow 1s} + B_{2p \rightarrow 1s} J_{\text{Ly}\alpha} + C_{2p \rightarrow 1s} n_e + M_{2p2s} + \Gamma_{2p}] n_{2p} - M_{2s2p} n_{2s} = (B_{1s \rightarrow 2p} J_{\text{Ly}\alpha} + C_{1s \rightarrow 2p} n_e) n_{1s} + \alpha_{2p} n_e n_{\text{HII}} + \frac{3}{4} S_{\text{chem}}, \quad (1)$$

$$[C_{2s \rightarrow 1s} n_e + M_{2s2p} + \Gamma_{2s} + A_{2s \rightarrow 1s}] n_{2s} - M_{2p2s} n_{2p} = C_{1s \rightarrow 2s} n_e n_{1s} + \alpha_{2s} n_e n_{\text{HII}} + \frac{1}{4} S_{\text{chem}}, \quad (2)$$

where the recombination source is $\alpha n_e n_{\text{HII}}$ (the electron recombines onto a proton; n_e and n_{HII} differ at the base, where helium and metals supply the electrons), $M_{2s2p} [\text{s}^{-1}]$ is the ℓ -mixing rate by electrons and ions, $M_{2p2s} = (g_{2s}/g_{2p}) M_{2s2p}$, and S_{chem} is the chemical $n = 2$ production of a molecular run, the dissociative recombination of H_2^+ and HeH^+ ($k_5 n_{\text{H}_2^+} n_e + k_{16} n_{\text{HeH}^+} n_e$, Koskinen et al. 2022 R5 and R16), shared between $2s$ and $2p$ by statistical weight. This is a 2×2 linear system in (n_{2p}, n_{2s}) ,

$$\begin{pmatrix} L_{2p} & -M_{2s2p} \\ -M_{2p2s} & L_{2s} \end{pmatrix} \begin{pmatrix} n_{2p} \\ n_{2s} \end{pmatrix} = \begin{pmatrix} S_{2p} \\ S_{2s} \end{pmatrix},$$

solved per radial cell by Cramer's rule. The $2s$ and $2p$ densities are carried separately into the Balmer opacity, whose sub-level oscillator strengths differ.

2.1 Rate coefficients

Every coefficient is the one the solver's level balance uses, transcribed in exhale_transit_lib.py from hydrogen_n2_rates.f90 and Cool_coeff.f90 (the Fortran name is given at each function, and the two agree to 3×10^{-13} relative over 10^3 – 10^5 K, the ℓ -mixing rates also over $n = 1$ – 10^{15} cm^{-3} , measured against the production objects):

Coefficient	Source and form
α_B	the case-B coefficient of the ionization balance: Badnell (2006, 2023 update) radiative recombination minus the ground capture of the Milne relation on the hydrogenic cross section (or the Koskinen et al. 2022 or Hui & Gnedin 1997 fit, as Atomic rate set and Legacy_HHe_rates of the run's input.inp select)
α_{2s}, α_{2p}	$f_{2s}(T) \alpha_B$ and $\alpha_B - \alpha_{2s}$, with f_{2s} the case-B fraction of Pengelly (1964, Table I); 0.322 at 10^4 K
$C_{1s \rightarrow 2s}, C_{1s \rightarrow 2p}$	$C \Upsilon(T) e^{-E_{12}/kT} / (2\sqrt{T})$, $C = h^2 / [(2\pi m_e)^{3/2} k^{1/2}]$, the CHIANTI v11.0.2 h_1 effective collision strengths (the rates of the H I cooling)
$C_{2s \rightarrow 1s}, C_{2p \rightarrow 1s}$	their detailed-balance reverses, Boltzmann factor cancelled analytically
M_{2s2p}	$n_e q^e + n_{\text{H II}} q^p + n_{\text{He II}} q^{\text{He}^+} + n_{\text{He III}} q^{\text{He}^{2+}}$: electrons from Seaton (1955, eq. 55, approximation V; 5.78×10^{-5} at 10^4 K), ions from the impact-parameter cross section of Pengelly & Seaton (1964) with their cut-off (protons 4.8×10^{-4} at 10^4 K)

The rates of Christie et al. (2013, Table 2) used here before 2026-09-25 differed in four places: $A_{2s \rightarrow 1s} = 8.26 \text{ s}^{-1}$; the Draine case-B fit with a quadratic f_{2s} ; power-law collision rates; and the electron ℓ -mixing rate of their R5 (5.35×10^{-6} at 10^4 K) with no ion term. Only the last moves the populations measurably: on the HD 209458 b _adv profile the proton term brings n_{2s}/n_2 from 0.53–0.69 down to 0.26–0.34 between 1.3 and $3 R_p$, the peak H α depth falls by 1.7 per cent and H β is unchanged; on the solved wasp_full state H α rises by 0.14 per cent (measured, md/stale_texts_20260925.md).

2.2 Ly α pumping and the $J_{\text{Ly}\alpha}$ input

The $1s \rightarrow 2p$ pumping rate is $B_{1s \rightarrow 2p} J_{\text{Ly}\alpha}$, with the Einstein coefficient in the mean-intensity convention

$$B_{1s \rightarrow 2p} = \frac{g_{2p}}{g_{1s}} \frac{c^2}{2h\nu_{\text{Ly}\alpha}^3} A_{2p \rightarrow 1s} \simeq 8.55 \times 10^9 \text{ cgs},$$

and the stimulated-emission rate $B_{2p \rightarrow 1s} J_{\text{Ly}\alpha}$ with $B_{2p \rightarrow 1s} = (g_{1s}/g_{2p}) B_{1s \rightarrow 2p}$. Here $J_{\text{Ly}\alpha}$ is the **angle-averaged mean intensity J_ν at Ly α line center**, in cgs units $\text{erg s}^{-1} \text{ cm}^{-2} \text{ Hz}^{-1} \text{ sr}^{-1}$, so that $B_{1s \rightarrow 2p} J_{\text{Ly}\alpha}$ has units of s^{-1} .

$J_{\text{Ly}\alpha}(r)$ is obtained in one of two ways.

(1) From a file. If Jlya_file points to an existing two-column text file, $J_{\text{Ly}\alpha}(r)$ is read from it:

column 1	r/R_p	(same radial coordinate as the EXHALE output)
column 2	$J_{\text{Ly}\alpha}$	[$\text{erg s}^{-1} \text{ cm}^{-2} \text{ Hz}^{-1} \text{ sr}^{-1}$]

linearly interpolated onto the EXHALE grid and clamped to the endpoints. A template ships as inputdata/Jlya.txt.example.

(2) Simple estimate (default; Huang et al. 2017). If no file is given, $J_{\text{Ly}\alpha}$ is estimated following Huang, Arras, Christie & Li (2017, ApJ 851, 150), Eq. (6) and the surrounding text,

$$J_{\text{Ly}\alpha}(r) \simeq 0.1 \frac{F_{\text{LyC}}}{\Delta\nu_D(r)}, \quad \Delta\nu_D = \nu_{\text{Ly}\alpha} \frac{1}{c} \sqrt{\frac{2k_B T}{m_p}},$$

where $\Delta\nu_D$ is the local Ly α Doppler width and F_{LyC} is the *deposited* (absorbed) Lyman-continuum energy flux. The prefactor encodes the assumption that each LyC photoionization is balanced by a recombination producing one Ly α photon; Huang et al. identify F_{LyC} with the total LyC flux absorbed inside the atmosphere ($2.6 \times 10^4 \text{ erg cm}^{-2} \text{ s}^{-1}$ for HD 189733b, page 11).

F_{LyC} from the actual stellar input. Rather than a fixed number, EXHALE_transit.py computes the absorbed LyC flux from the EXHALE inputs:

$$F_{\text{LyC}} = \xi \underbrace{\frac{10^{L_{\text{EUV}}}}{4\pi a^2}}_{\text{incident stellar LyC}} \times \underbrace{(1 - e^{-\tau_{\text{LyC}}})}_{\text{fraction absorbed}}, \quad \tau_{\text{LyC}} = \sigma_{\text{LyC}} N_{\text{HI}},$$

where L_{EUV} (the EUV-band luminosity, $E > 13.6 \text{ eV}$) and the orbital distance a are read from input.inp, N_{HI} is the vertical neutral-hydrogen column from the EXHALE profile, and $\sigma_{\text{LyC}} \simeq 6.3 \times 10^{-18} \text{ cm}^2$. The factor ξ is the day-night / 2D flux dilution, taken to match ATES's "2D approximate method" (Rate/2 $\rightarrow$ 0.5, Rate/4 $\rightarrow$ 0.25, else 1; read from input.inp, cf. the ξ factor of Christie+2013 / Huang+2017). For the optically-thick atomic layer $\tau_{\text{LyC}} \gg 1$, so the absorbed fraction $\rightarrow 1$ and $F_{\text{LyC}} \rightarrow \xi \times$ the incident stellar LyC flux (e.g. $\xi = 0.5$, $\simeq 5.2 \times 10^2 \text{ erg cm}^{-2} \text{ s}^{-1}$ for the HD 209458b input). A fixed value can be forced via F_LyC_override and the dilution via xi_override.

This estimate is a uniform-illumination approximation: it omits the decline of $J_{\text{Ly}\alpha}$ deep in the atmosphere captured by the full radiative-transfer treatment (Huang et al. Eq. 7) and the direct stellar Ly α component; for a self-consistent profile, supply Jlya_file.

2.3 $n = 2$ photoionization Γ_{2s}, Γ_{2p} (Balmer continuum)

The $n = 2$ levels are photoionized by the stellar Balmer continuum ($E > 3.4 \text{ eV}$, $\lambda < 3647 \text{ \AA}$); this depletes n_2 and is included in Eqs. (12)–(13) through Γ_{2s}, Γ_{2p} . The solver integrates the Balmer band of the run's own spectrum type, but neither that spectrum nor the rate is in the profiles this tool reads, so here the field is a diluted stellar blackbody: for a stellar effective temperature T_{star} and radius R_{star} ,

$$\Gamma_2 = \int_{\nu_2}^{\nu_1} \frac{F_\nu}{h\nu} \sigma_2(\nu) d\nu, \quad F_\nu = \pi B_\nu(T_\star) \left(\frac{R_\star}{a}\right)^2, \quad \sigma_2(\nu) = \sigma_{2,\text{th}} \left(\frac{\nu_2}{\nu}\right)^3,$$

over the band the solver uses, from the $n = 2$ edge $h\nu_2 = 3.3996 \text{ eV}$ to the H I edge $h\nu_1 = 13.5984 \text{ eV}$ (above it the ground state absorbs the photons), with $\sigma_{2,\text{th}} = 1.4 \times 10^{-17} \text{ cm}^2$ for both levels and $\Gamma_{2s} = \Gamma_{2p} = \Gamma_2$. The solver states no source for $\sigma_{2,\text{th}}$; it is, to its two figures, the $n = 2$ threshold value of the Kramers cross section with the bound-free Gaunt factor of Seaton (1959, MNRAS 119, 81, eqs. 3 and 10), $1.378 \times 10^{-17} \text{ cm}^2$, and the exact hydrogenic threshold values are 1.478×10^{-17} ($2s$) and 1.355×10^{-17} ($2p$) cm^2 , statistical mean $1.386 \times 10^{-17} \text{ cm}^2$ (derived by direct integration of the bound-free dipole matrix elements). This reproduces Huang et al.'s Table 2 values ($\sim 20\text{--}26 \text{ s}^{-1}$) for a K dwarf, and is substantially larger for a hotter G dwarf (e.g. $\sim 140 \text{ s}^{-1}$ for HD 209458's $T_\star \simeq 6065 \text{ K}$), which suppresses n_2 and hence the Balmer-line depth. Setting $T_{\text{star}} \leq 0$ falls back to the manual constants Gamma_2s/Gamma_2p (default 0).

3. H α and H β optical depth and transmission

Once $n_2(r)$ is known, the H α transmission is computed with the same LOS integration as Ly α . Along a chord at impact parameter b ,

$$\tau_\nu(b) = \int n_2(r) \sigma_{\text{H}\alpha}(\nu, r) d\ell, \quad \sigma_{\text{H}\alpha} = \frac{\sqrt{\pi} e^2}{4\pi\epsilon_0 m_e c} \frac{f_{23}}{\Delta\nu_D} H(a, x),$$

where $H(a, x) = \text{Re}[w(x + ia)]$ is the Voigt function (Faddeeva w), $\Delta\nu_D = \nu_{\text{H}\alpha} v_{\text{th}}/c$ uses the proton thermal speed, and the line-of-sight bulk velocity is folded into x exactly as for the existing lines. Atomic data: $\lambda_{\text{H}\alpha} = 6562.8 \text{ \AA}$ (air), $f_{23} = 0.6407$, $A_{3 \rightarrow 2} = 4.41 \times 10^7 \text{ s}^{-1}$ (the line is strongly Doppler-dominated, so the damping wing is negligible). The disk-averaged transmission, instrument convolution and planet-rotation convolution reuse the existing code paths.

H β ($n = 2 \rightarrow n = 4$). The same $n = 2$ population also produces H β at $\lambda_{\text{H}\beta} = 4861.35 \text{ \AA}$ ($f_{24} = 0.119$, $A_{4 \rightarrow 2} = 8.42 \times 10^6 \text{ s}^{-1}$). It is computed with an identical LOS loop and plotted/printed alongside H α . Because both lines are optically thick (saturated), the H α /H β depth ratio is much smaller than the optical-depth ratio (~ 7.3): e.g. on the converged HD 209458b run, H $\alpha \simeq 3.2\%$ and H $\beta \simeq 2.0\%$ with TPM's default star, and $\simeq 0.41\% / 0.26\%$ with HD 209458's actual $R_\star, T_\star$ (see the comparison notebook).

4. Implementation in EXHALE_transit.py

System parameters. The planet and the star are taken from the run directory, matched by label exactly as `input_read.f90` does. Besides R_p , M_p , T_{eq} , a , $M_\star$, L_{EUV} and the 2-D approximate method, this includes the stellar radius (`Stellar radius [R_sun]`) and effective temperature (`Stellar Teff [K]`), which set the transit normalization $A_\star$, the $R_\star/R_p$ cap on the absorbing annulus, and the diluted-blackbody Balmer continuum of §2.3. The planet spin period has no `input.inp` key: a close-in giant is tidally locked, so it defaults to the Keplerian orbital period $P_{\text{orb}} = 2\pi\sqrt{a^3/[G(M_\star + M_p)]}$. Each of the three is resolved as `EXHALE_TRANSIT_RSTAR_RSUN / _TSTAR / _ROTP` environment override `> ./input.inp >` built-in default, and the resolved values and their sources are printed at startup.

EXHALE_resolved.out comes first (2026-08-22). $R_p, M_p, T_{\text{eq}}, a, M_\star$ and He/H are read from `EXHALE_resolved.out` beside `input.inp` whenever that file exists, and only from `input.inp` when it does not. The wind solver writes it (`write_resolved_config` in `write_setup_report.f90`) after a `base.inp` handoff or a lower-atmosphere profile has moved the base, so it is the geometry the wind actually used, while `input.inp` alone is stale in exactly that case. `read_input_params` says at startup which of the two it took, and a file that is present but unreadable raises rather than falling back silently. The stellar radius and T_{eff} are not in that file and keep the precedence above.

- `n2_populations(T, n1s, nHII, ne, Jlya, G2s, G2p)` solves the 2×2 system above and returns (n_{2s}, n_{2p}, n_2) in cm^{-3} . The Ly α -pump constants $B_{1s \rightarrow 2p}$, $B_{2p \rightarrow 1s}$ are precomputed.
- $J_{\text{Ly}\alpha}$ is read from `Jlya_file` if present, otherwise built from the Huang et al. (2017) estimate (§2.2); n_2 is then built into the symmetric (inverted+normal) radial profile `data_n2` (in m^{-3}), exactly like `data_nHI`.

- A new wavelength grid `l_nde_Ha` and an H α LOS loop fill `exp_tau_Ha`; the disk average, instrument convolution (`Instr_res_Ha`) and rotation convolution produce the plotted curves.
- The H α branch always runs (`do_Ha=True`); with the default `Jlya_file=''` it uses the Huang et al. (2017) estimate for $J_{Ly\alpha}$, with F_{LyC} taken as the absorbed stellar LyC flux (§2.2).
- **Compatibility fix.** `Ion_species.txt` in EXHALE carries trace-metal columns, so the reader was changed to `usecols=range(7)` to take only r and the six H/He species. This also remains correct for the old 13-column (C/O-only) and 7-column (H/He-only) outputs. The column count has since grown: `write_output.f90` writes r , the six H/He species (HI HII HeI HeII HeIII HeITR) and all `n_mion = 27` metal-ion stages, i.e. **34** columns, and **38** on a molecular run, which appends H2 H2p H3p HeHp; the metal columns are written whether or not metals are on. The `usecols=range(7)` slice is unaffected (the first seven columns are still r plus the six H/He species), and anything past them is found through the file's own `# columns` schema header.

4.1 Outputs

Every line carries one key (He10830, Lya, Halpha, Hbeta, MgII, CaII, NaI), and both products of a line are named from that key and land in the run directory:

Product	Name	Default
model curve	<code><path>/tpm_<line>.txt</code>	always written
figure	<code><path>/<prefix><line>.png (+ .pdf)</code>	off; set EXHALE_TRANSIT_FIG_PREFIX

A curve file has the columns `lambda[A]`, T_λ (theoretical), T_λ (instrument-convolved), T_λ (+rotation), so the excess absorption in percent is $(1 - T_\lambda) \times 100$. The set follows the run: a metals-off run writes no metal curves. EXHALE_TRANSIT_SAVE_PREFIX decorates the curve name exactly as the figure prefix decorates the figure name, and both are resolved against the run directory, so an absolute prefix writes elsewhere and nothing depends on the working directory. Every curve file also opens with a block of `#` comment lines stating what the spectrum was built on: the input profile and its validity schema, the census of the rows the chords sampled, and the overrides in effect (§5.3). They are comments, so the four numerical columns are what they always were.

5. What the spectrum is built on: row validity, the census, and the metadata block

The spectra are integrated over the `*_adv.txt` profiles, and those profiles do not carry one uniform product: the steady advective correction is applied to some rows and refused on others. Since the profiles began stating that row by row, the transit tool reads the statement, reports what the chords sampled, and writes both into every curve it saves. None of this changes a numerical column of a `tpm_*.txt` file.

5.1 Reading the profile's own statement of validity

`read_adv_validity(exhale_transit_lib.py)` decides which of three cases a profile is, by its header and by nothing else. Columns are located by name on the `# columns` line, so a file that gains or reorders columns is still read correctly.

adv_schema	What the file carries	Row decision
2	adv_T_status, adv_comp_status, adv_mass_row	refused when adv_T_status $\neq 0$ or adv_comp_status = 2
1	one adv_status column	refused when it is nonzero
absent	no schema line and no status column	every row UNKNOWN

Under schema 2 each status field takes one of five values, 0 corrected, 1 retained, 2 failed, 3 unsupported, 4 not_evaluated. The census refuses a row whose temperature field is not zero, or whose composition field is failed: a corrected temperature on a composition whose solve did not converge is not a corrected row (refused_reason is 2 when either field is failed, else the temperature field). A retained composition under a corrected temperature is not refused; the composition field is reported beside the census, because the lower-level density of every line comes from that composition. A refused field of the profile carries the run's own numbers exactly, except the temperature of a failed row of a rejected column energy solve, which is the marching profile of the post-process; such a row exists only in a profile whose record is rejected (below).

A corrected row is a *conditional* correction, not a certified one. The profile states the fraction it is accurate to in the mass flux, `adv_conditional_tol` (currently 10^{-2} , the first-order argument at its declaration), and the measure each row was decided by, `adv_mass_row` = $|R_1|/s_1$, the mass row of the certification's own residual scaled by that row's own largest term. The header's `adv_input_certified` line is a different statement and is kept separate: it is the certification's verdict on the run state, at the much tighter `cert_tol_mass`, not a verdict on the post-processed rows. The reader returns the tolerance, its header text and the measure column, so the spectrum can carry the condition of the rows it stands on.

The derived-state record. Beside the row fields the profile carries one `# adv_derived_state:` line, the record of how the solves that produced the rows ended as a whole (the column energy solve of the transport terms, the chemistry, the outer iteration), read back as the dict `derived_state` and judged by `adv_derived_state_verdict`: rejected, unverified (the best a record says: the outer iteration is a fixed number of passes and the column is solved with lagged coefficients), or unknown when the file has no such line. A rejected or unknown profile gives no spectrum: EXHALE_transit.py exits with status 7, the status the evaluation itself exits with for a rejected product, unless EXHALE_TRANSIT_DIAGNOSTIC=1, which makes the curves as diagnostic data. The solution selection has no such record and is not judged by one.

A legacy profile is read as UNKNOWN and said to be UNKNOWN. It is not read as "corrected": a writer that did not state its refusals says nothing about them, and a census of such a file can only say so. Both the console and the saved metadata print that distinction explicitly, because UNKNOWN is not zero refused.

5.2 The census: which rows the chords sample

A ray at impact parameter b crosses every shell with $r \geq b$, at line-of-sight coordinate $|x| = \sqrt{r^2 - b^2}$, and there is *no upper radial limit*: a shell at $r = 12$ meets the ray at $b = 2$ whether or not the ray grid stops at $b = 10$, so a shell lying outside the stellar disk still absorbs on the rays that cross the disk. One function, `chord_shell_indices`, now serves the census, `resonance_depth` and `resonance_spectrum` alike; two selections would let the diagnostic describe a different atmosphere from the one the spectrum was built from. The earlier census mask imposed an upper limit at the stellar-disk cap R_{ib} and therefore described less material than the spectrum contains. That defect is latent whenever R_{ib} is the outermost grid radius, which both regression profiles are, and real as soon as the domain passes the stellar radius: MEASURED on `wasp_full` read at `EXHALE_TRANSIT_RSTAR_RSUN=0.30`, the sampled rows go from 419 to 500, 81 of them above the cap.

The console reports, and the metadata block records, how many of the sampled rows are not the steady advective correction, broken down by reason and with the composition field beside it. Then, line by line at eight impact parameters and at its maximum, it reports the share of the line-center optical depth carried by those rows. The share is exact rather than estimated: the optical depth of a ray is the trapezoid sum $\tau(b) = \sum_k w_k I_k$ over the rows the chord crosses, which is additive, so a subset of the rows carries exactly $\sum_{k \in \text{subset}} w_k I_k / \tau(b)$ (`refused_line_center_tau_share`).

It is a CONTRIBUTION, not an uncertainty. The disk-averaged transmission is an average of $\exp(-\tau)$ and is nonlinear in τ , so a saturated ray moves very little when its optical depth moves, and a share near one does not mean the depth is wrong by a comparable amount. The tool labels it as such in the console and in the file, and it should be quoted that way. On `wasp_full` the rays near $b = 1$ take 99 percent of their Ly α line-center depth from uncorrected rows (MEASURED), which says where the material is, not that the depth is unreliable.

5.3 The metadata block on every saved curve

`transit_metadata_block` writes a block of # comment lines at the head of every `tpm_*.txt` product, the two handwritten products included. Comments only: `np.loadtxt` and any reader that skips comment lines sees exactly the numerical columns it saw before, and the numerical columns were verified bitwise across all ten `wasp_full` and five `mol_base_handoff` products when the block was introduced. The block carries, in order:

- `transit_schema` (currently 1) and a one-line statement that the product is a one-way transmission of the profile named next;
- the input profile's path, its `adv_schema`, its `adv_input_certified` verdict, its stationarity operator, its `adv_conditional_tol` text and the largest `adv_mass_row` over its rows, and its model restrictions, all copied out of the profile's own header, so a reader of the curve alone can tell what it stands on;
- for an `_adv` profile, its derived-state record (`adv_derived_state`), and in the diagnostic mode an `adv_derived_state_refusal` or `adv_derived_state_unknown` line saying that the curve is diagnostic data and not a result;
- the profile's provenance lines;

- the tool’s own identity, by git hash and file modification times;
- the census for this line: sampled rows, refused rows, rows above the disk cap, the refusal reasons, the composition breakdown, the eight-point refused-share ladder and its maximum with the b it occurs at, and the sentence saying the share is a contribution and not an uncertainty;
- every EXHALE_TRANSIT_* or TPM_* override in effect, or a line saying that none is and every window, resolving power and geometry setting is the built-in default or `input.inp`.

When the input states no row validity the census line says so, in the words “the count is UNKNOWN, which is not zero refused”.

6. Validation

Run end-to-end on the converged HD 209458b EXHALE output (`output/*_adv.txt`), the module is stable and NaN-free and yields clean H α and H β absorption lines at 6562.8 and 4861.4 Å. With the Huang $J_{\text{Ly}\alpha}$ estimate ($\xi = 0.5$, $F_{\text{LyC}} \simeq 5.2 \times 10^2$, Γ_2 from $T_\star = 6065$ K), the depths are physically reasonable and the H α /H β ratio reflects saturation (both optically thick). The Ly α branch in the same run saturates ($\sim 100\%$ core absorption) and He I behaves as before. A worked comparison against the Jensen et al. (2012) HD 209458b observation is provided in the notebook `Halpna_compare_HD209458b.ipynb`.

7. Metal resonance doublets (Mg II h&k, Ca II H&K, Na I D)

The metal resonance lines are the key NUV/optical diagnostics of metal-enriched escaping atmospheres (Huang et al. 2023). They absorb out of the *ion ground state*, whose excited fine-structure levels are Boltzmann-negligible at $\sim 10^4$ K, so the lower-level density is the ion density itself ($n_{\text{lower}} \simeq n_{\text{ion}}$): no non-LTE population calculation is needed. The ion-density profiles are read from the metal block of `Ion_species_adv.txt`; a metals-off run (all-zero columns) skips this section automatically.

7.1 Atomic data and implementation

Each species is treated as a **doublet**, with both components summed in a single wavelength window (Table 1; NIST f -values and A -coefficients). The optical depth is computed exactly as for H/He: spherical chords $\sqrt{r^2 - b^2}$ on the impact-parameter grid, LOS velocity $v_{\text{los}} = vx/r$, Voigt profiles via the Faddeeva function (vectorized over wavelength), trapezoidal τ , projected-area disk average, then the instrument convolution and the planet-rotation convolution with the depth-dependent effective radius $R_{\text{eff}} = R_p \sqrt{(h + \delta)/\delta}$ (capped at the impact-parameter boundary), $\delta = (R_p/R_\star)^2$. The three doublets are written under the same `tpm_<line>` naming as the H/He lines (§4.1), with the keys MgII, CaII, NaI. Alongside the spectra, the script prints the validated *single-component* depth table (line-center and 4 Å-band percentages for Mg II 2796, Ca II K, Na I D2), which is the quantity used for the Huang et al. (2023) comparisons; the doublet peak differs from the single-component value only by the partner-line wing (e.g. 3.75 vs. 3.72% for Mg II

Table 1: Metal resonance doublets in EXHALE_transit.py (NIST).

species	component	λ_0 [Å]	f	A_{21} [s $^{-1}$]	window / R_{instr}
Mg II	k	2796.352	0.608	2.60×10^8	2790–2810 Å / 3×10^4
	h	2803.531	0.303	2.57×10^8	
Ca II	K	3933.663	0.6267	1.47×10^8	3927–3975 Å / Instr_res_Ha
	H	3968.469	0.3116	1.40×10^8	
Na I	D2	5889.951	0.641	6.16×10^7	5884–5902 Å / Instr_res_Ha
	D1	5895.924	0.320	6.14×10^7	

on WASP-121b). Two conventions matter when comparing with Huang: their table lists the effective transit radius $R_{\text{eff}}/R_* = \sqrt{(R_p/R_*)^2 + h}$, not a percent absorption, and Mg II is quoted in a 4 Å NUV bin while the optical lines are line-center (line-center vs. 4 Å differ by $\sim 3\times$ for the optically thick Mg II core).

For a Roche-lobe run the 3-D equipotential reconstruction (geometry = 'triaxial', roche_recon.py) replaces the spherical chords by the full Roche geometry with an LOS velocity for each sector $v_{\text{sub}}(r_{\text{eff}})x/r - \Omega y$ (wind + tidally locked rotation) and reports the same depth table.

7.2 Example spectra

Figure 1 shows the three doublet spectra for the WASP-121b worked example (backup/regression/tpm_wasp/, Roche Newton solution, metals on). The optically thick cores saturate, so the two doublet members reach similar depths despite the 2:1 oscillator-strength ratio; the rotation convolution mainly shallows and broadens the narrow Na I D cores.

8. Caveats

1. **$n = 2$ photoionization.** Γ_{2s}, Γ_{2p} are now estimated from a diluted stellar-blackbody Balmer continuum (§2.3, parameter T_star); this is a single blackbody approximation to the true stellar near-UV spectrum. Setting T_star ≤ 0 reverts to the manual constants (default 0).
2. **Electron density.** $n_e = n_{\text{HII}} + n_{\text{HeII}} + 2n_{\text{HeIII}}$ from EXHALE is used (Christie assume $n_e = n_p$); the difference is small.
3. **$J_{\text{Ly}\alpha}$ convention.** The input must be the mean intensity J_ν at line center in cgs (§2.2). A different convention (energy density, frequency-integrated, etc.) requires rescaling the pump constant.
4. **Wavelength/damping.** $\lambda_{\text{H}\alpha}$ is the air value 6562.8 Å; the natural width is negligible relative to the thermal Doppler width, so the line is essentially Gaussian.

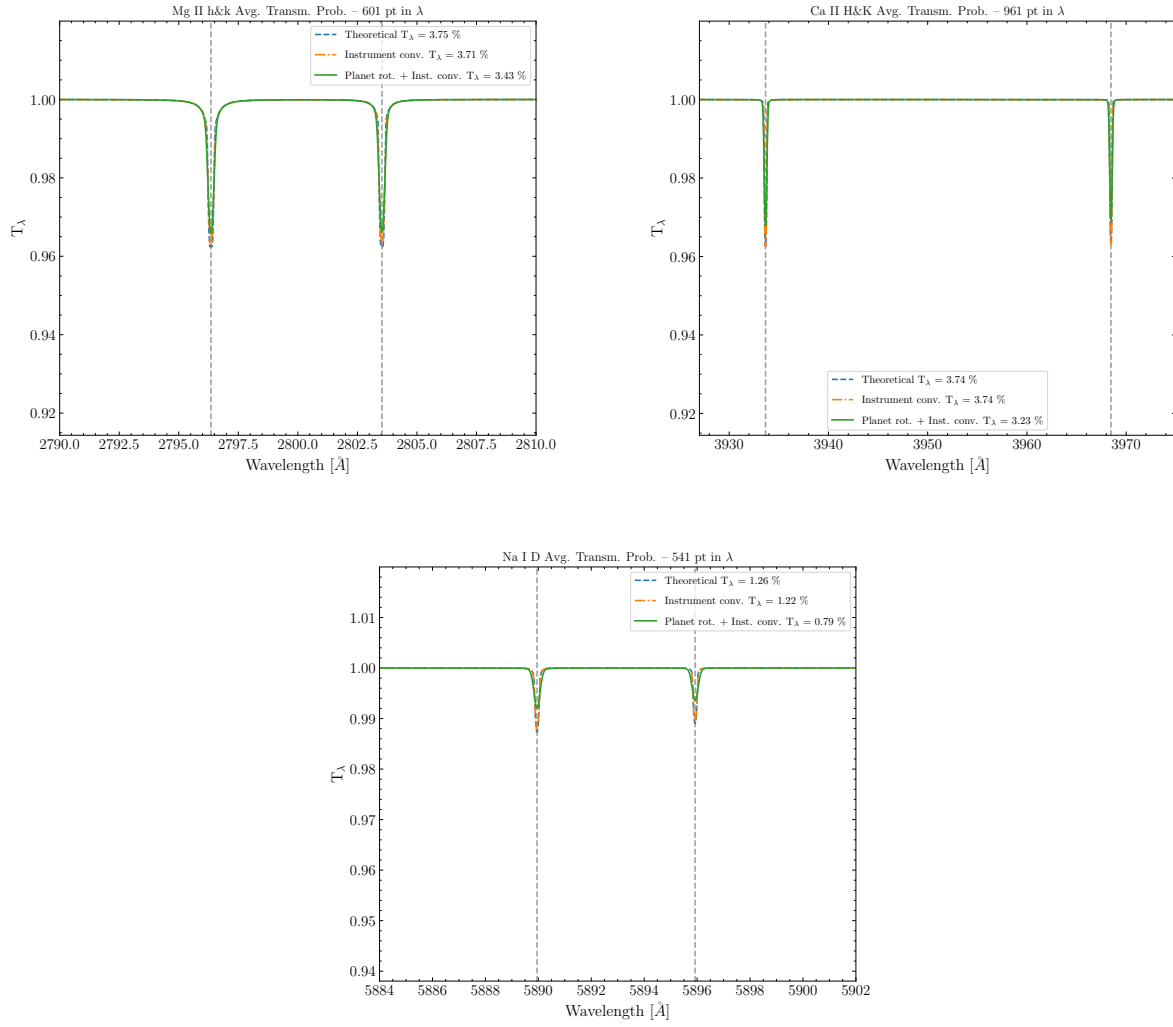


Figure 1: Metal resonance doublet transmission spectra for WASP-121b (backup/regression/tpm_wasp/): Mg II h&k (top left), Ca II H&K (top right), and Na I D (bottom). Each panel shows the theoretical, instrument-convolved, and instrument+rotation-convolved transmission, as for the H/He lines.

Reference

- Christie, D., Arras, P., & Li, Z.-Y. 2013, ApJ, 772, 144: “ $H\alpha$ Absorption in Transiting Exoplanet Atmospheres” (references/Christie_2013ApJ_772_144.pdf).
- Huang, C., Arras, P., Christie, D., & Li, Z.-Y. 2017, ApJ, 851, 150: “A Model of the $H\alpha$ and Na Transmission Spectrum of HD 189733b” (references/Huang_2017_ApJ_851_150.pdf); the $J_{Ly\alpha}$ estimate is their Eq. (6) and related text.
- Jensen, A. G., et al. 2012, ApJ, 751, 86: transit $H\alpha$ spectroscopy of HD 209458b and HD 189733b (comparison target for the HD 209458b notebook).